

JUHI DAMLEY

jdamley28@cmc.edu | (305) 538-2856 | San Francisco, CA | [GitHub](#) | [Portfolio](#) | [LinkedIn](#)

EDUCATION

Computer Science junior building AI systems at the intersection of political science and stochastic optimization

Claremont McKenna College / Harvey Mudd College

Claremont, CA

Bachelor of Arts, Computer Science (cross-campus major at HMC)

August 2024 – May 2028

Activities: Girls Who Code (Founder & President), Sagehen Capital Management, Board of Trustees Advancement Committee, p-AI, HMC Grader/Tutor for CS5 and CS70

Coursework: Digital Electronics & Computer Engineering, Stochastic Operations Research, Abstract Algebra, Linear Algebra, Statistics, Data Structures,

EXPERIENCE

Electoral Equilibrium | Researcher

May 2026 – Present

Claremont McKenna College Summer Research Program- Supervised by Prof. Gaston Espinosa

- Built a stochastic optimization framework that ingests a hypothetical political shock and prescriptively outputs how a voter coalition must rebalance across Race $\times$ Religion $\times$ Gender strata to maintain a win, with a Monte Carlo 90% confidence interval on win probability
- Fine-tuned Mistral 7B via QLoRA on CMC HPC A100s to classify demographic loyalty shifts, deployed via Modal with constrained decoding and SSE streaming to a Next.js frontend
- Built a distributed pipeline ingesting 2B+ messages across Discord, Reddit, Bluesky, Telegram, and Twitter archives; deployed a 3-stage sequential bio-classifier (keyword matching $\rightarrow$ SetFit $\rightarrow$ language priors) for demographic inference

The Gould Center for Humanistic Studies | Research Assistant

October 2024 – Present

- Delivered quantitative research for Prof. Espinosa's "Religion, Race, Gender, and the 2024 Election" series; produced 70+ data visualizations for conferences, a five-book series, and an upcoming podcast
- Built full-stack research website in TypeScript/React/Supabase; assessed project viability via web scraping and OpenAI API; managed logistics for multi-speaker national conference

Financial Economics Institute | Research Analyst

January 2026 – May 2026

- Built config-driven Python portfolio optimization pipeline: CRSP panel ingestion, deterministic run keys, typed JSON/Parquet artifacts, LedoitWolf-regularized covariance estimation on CMC HPC
- Applied Markowitz mean-variance optimization to large asset universes; identified min-variance objective limitations that directly motivated Electoral Equilibrium's DQCP quasi-convex formulation

Claremont McKenna College | Office of Advancement Intern

October 2024 – December 2025

- Advised Board of Trustees on AI integration; built Gemini classification model to automate categorization of 4,000+ alumni records, cutting processing time 50%+
- Managed alumni data pipeline: integrated third-party platforms, digitized archival files, standardized records across systems; built Python automation reducing manual updates by half

SKILLS

LANGUAGES	Python, C++, TypeScript, Java, R, Swift, Racket
AI / ML	PyTorch, HuggingFace Transformers, QLoRA, Mistral 7B, RoBERTa, SetFit, LLM Fine-Tuning, Constrained Decoding, Claude Code, Github Copilot, Prompt Engineering, LLM APIs
DATA	CVXPY, NumPy, SciPy, pandas, PostgreSQL, Supabase, Parquet, Monte Carlo Simulation
INFRASTRUCTURE	Modal, SLURM / HPC, Prefect, FastAPI, Next.js, Vercel, Firebase, Git
PRODUCT PROJECTS	Figma, Notion, Canva, Adobe Photoshop

- **LectureTeX** | lecturetex.juhi.studio Python, FastAPI, Modal — Lecture-to-PDF notetaking app: speech captured via Faster Whisper on Modal A100 GPUs flows into subject-aware L^AT_EX generation across 10 LLMs, automated compilation, and delivery via a DigitalOcean MCP server. FastAPI backend, React web and Electron desktop clients.
- **WakeUp** Swift — Accessibility app for narcolepsy: continuously monitors Apple Watch biosensor streams to detect excessive daytime sleepiness onset and triggers haptic wakefulness interventions in real time.
- **juhi.studio** | juhi.studio TypeScript — Personal portfolio with a live Mandelbrot fractal background, Conway's Game of Life animations, and a retro Windows 95 UI housing two integrated sub-applications.
- **Prioritize** | ptz.juhi.studio TypeScript — Priority-queue task manager with real-time collaborative features; lets teams organize and rank tasks by urgency with a clean drag-and-drop interface.
- **devlog** | devlog.juhi.studio TypeScript — Public build log documenting Electoral Equilibrium and other ongoing projects in real time; a running record of architectural decisions, pivots, and lessons learned.
- **Girls Who Code at the Claremont Colleges** | gwc5c.org TypeScript — Founded and built the 5C chapter of the national Girls Who Code organization in 2026; shipped the chapter website with a members-only portal for workshop resources and exclusive content.

HONORS & AWARDS

Handley Leadership Fellowship • National Merit Scholar • Venture Capital Program Finalist • CMC SP '25 Statistics Competition Winner • AHS Merit Scholarship (\$85,000)